# IF MACHIAVELLI HAD KNOWN DARWIN: EVOLUTIONARY FOUNDATIONS OF MODERN GOVERNANCE

Ignacio Adrián Lerer
Independent Researcher
Buenos Aires, Argentina
adrian@lerer.com.ar
ORCID: 0009-0007-6378-9749

November 2025

## ABSTRACT

Political theory cannot explain why manifestly dysfunctional institutions persist across generations despite repeated reform attempts. When Niccolò Machiavelli wrote *Il Principe* in 1513, he revolutionized political thought by describing power as it IS rather than as it SHOULD BE. But writing 346 years before Darwin published *On the Origin of Species*, he analyzed princes rather than institutions, personal cunning rather than evolutionary fitness. This paper completes Machiavelli's project by integrating Darwin's natural selection, Dawkins's extended phenotypes, and Dennett's memetics into a unified framework for governance. The core principle is stark: institutions optimize for memetic replication, not social welfare. This fitness-welfare divergence explains why "bad" institutions systematically outcompete "good" alternatives.

Empirical validation demonstrates the framework's predictive power. Argentine populism exhibits 216-fold reproductive advantage over liberalism despite chronic economic dysfunction. The Constitutional Lock-in Index (CLI) predicts reform success with 87% accuracy across multiple jurisdictions: Argentina's CLI of 0.89 correlates with zero sustained reforms across 23 attempts over 80 years, while Chile's CLI of 0.24 correlates with 83% bidirectional reform success. Legal transplants fail at 76% rates globally; environmental compatibility analysis predicts outcomes with 87.4% accuracy, vastly outperforming technical quality assessment. The WEIRD versus non-WEIRD implementation gap, a 5.7-fold disparity stable for four decades, remains unexplained by conventional theory but follows naturally from memetic alignment principles.

The paper makes four conceptual contributions. First, the fitness-welfare divergence principle explains institutional persistence independent of outcomes. Second,

Evolutionarily Stable Strategy (ESS) analysis clarifies why superior alternatives fail to displace inferior incumbents. Third, the ultraactivity mechanism reveals how temporal asymmetry transforms repeated games into terminal competition. Fourth, computational tools. IusMorfos for transplant prediction, CLI for institutional rigidity measurement, JurisRank for doctrine persistence analysis, operationalize evolutionary governance.

The implication is unequivocal: modern statecraft requires evolutionary epistemology. The Prince of the 21st century engineers institutional ecosystems, not personal power. This is not naturalistic fallacy, evolution does not justify governance. Rather, evolutionary analysis enables prediction of which reforms persist and which collapse, regardless of technical merit or moral worth. Where Machiavelli taught princes to acquire and maintain personal authority, this framework teaches governors to construct self-replicating institutional machinery that endures across generations.

**JEL Classification:** B52, D02, P16, K00, C73

**Keywords**: Machiavelli, Darwin, Dawkins, extended phenotype, memetics, evolutionarily stable strategies, constitutional lock-in, institutional persistence, governance, legal transplants, Argentina, comparative politics

## I. INTRODUCTION: WHY MACHIAVELLI NEEDS DARWIN

When Niccolò Machiavelli composed Il Principe during his forced retirement in 1513, he committed intellectual heresy that reverberates five centuries later. His crime was not immorality, though contemporaries accused him of teaching evil, but rather an unflinching commitment to describing political power as it actually operates, stripped of theological justification and moral decoration. Where medieval and Renaissance political philosophy had concerned itself primarily with how princes should govern according to divine law or natural justice, Machiavelli asked a fundamentally different question: how do princes actually acquire, maintain, and lose power? This methodological revolution, empirical observation over normative prescription, established the foundations of modern political science. Max Weber's value-free sociology, Vilfredo Pareto's analysis of elite circulation, and Joseph Schumpeter's economic theory of democracy all trace their intellectual lineage to Machiavelli's insistence that political analysis begin with facts rather than ideals.

Yet Machiavelli wrote 346 years before Charles Darwin published On the Origin of Species in 1859. He could not have known that the political dynamics he observed, the

rise and fall of princes, the persistence and collapse of states, the cycles of republic and tyranny, obey the same fundamental principles governing biological evolution: variation, selection, differential reproduction, and adaptation. Machiavelli's conceptual vocabulary centered on virtù, the skill, courage, and cunning enabling a prince to master circumstance, and fortuna, the capricious external forces that no amount of virtue could fully control. These concepts captured important aspects of political reality, but they remained bound to individual agency and historical contingency. Machiavelli analyzed princes, individual rulers whose personal qualities determined state survival, because he lacked a framework for understanding how institutional structures themselves could act as autonomous replicators, persisting across rulers and even against rulers' intentions.

The gap between Machiavellian political analysis and evolutionary biology is not merely historical but conceptual. Machiavelli's Discorsi analyzed republican institutions with sophistication, recognizing that certain constitutional arrangements proved more durable than others. His examination of Roman republican mixed government demonstrated clear understanding that institutional design shapes political outcomes across generations. Yet even this institutional analysis remained grounded in intentional design, legislators consciously crafting constitutions, citizens deliberately maintaining civic virtue. The possibility that institutions themselves might evolve through selection pressures, accumulating features that maximize their own replication rather than collective welfare, lay beyond Machiavelli's conceptual horizon. He understood that bad institutions could persist through inertia, corruption, or external support, but he lacked the theoretical apparatus to explain why dysfunctional institutional arrangements systematically outcompete functional alternatives across centuries.

Modern political theory has advanced beyond Machiavelli in many dimensions, democratic theory, institutional economics, game-theoretic analysis of strategic interaction, yet it paradoxically remains pre-Darwinian in a crucial respect. Contemporary political science excels at explaining how institutions function, how actors respond to incentives, and how coalitions form and dissolve. What it cannot explain, except through ad hoc historical narration, is why manifestly dysfunctional institutions persist across generations despite repeated reform attempts, substantial resources invested in institutional capacity building, and widespread recognition of dysfunction. Why did no Argentine liberal government complete a full eight-to-ten-year political cycle from 1946 through 2025, despite six attempts and strong electoral mandates? Why do legal transplants, taking a law that functions well in one jurisdiction and implementing it identically in another, fail at 76% rates globally, even when both

source and recipient share legal traditions, language, and economic development levels? Why does the implementation gap between law-on-books and law-in-action in non-WEIRD societies remain 5.7 times larger than in WEIRD societies, stable across four decades and resilient to massive investments in capacity building?

These are not mere historical curiosities requiring narrative explanation but systematic patterns demanding theoretical account. Rational choice institutionalism predicts that inefficient institutions should be replaced by efficient ones, yet institutionally locked dysfunction persists. Historical institutionalism invokes path dependency to describe institutional stickiness, but "path dependency" is description masquerading as explanation: it tells us that institutions persist but not why or under what conditions they become vulnerable to displacement. Behavioral economics illuminates cognitive biases affecting individual decision-making but cannot explain multi-generational institutional crystallization that long outlives any individual's cognitive limitations. Constructivist approaches emphasize that institutions are socially constructed and therefore potentially reconstructible, yet this theoretical flexibility produces empirical brittleness, if institutions are mere constructions, why do certain constructions prove so resistant to deconstruction despite concerted effort?

The explanatory gap becomes particularly acute when we observe patterns that seem to defy rationality altogether. Consider Argentina's tax system. The National Constitution of 1853, following the United States federal model, explicitly requires that national direct taxes be established "for determinate time" (Article 75, clause 2). Yet the income tax created by Law 11,682 in 1932 has operated continuously for 93 years without temporal limitation and without constitutional amendment authorizing permanent national income taxation. Argentina remains the only federal case globally where national income tax operates without explicit constitutional authorization, compare the United States' Sixteenth Amendment (1913), Australia's referendums (1910-1967), or India's constitutional provisions (1950). This is not ignorance or oversight; Argentine constitutional scholars have documented the irregularity for decades. It is not impossibility, the Constitution was successfully amended in 1949, 1957, and 1994 on other matters. Yet this particular constitutional violation persists because any attempt to regularize it confronts an impossible trilemma: Argentina can simultaneously achieve only two of three objectives, constitutional compliance, fiscal sustainability, and provincial autonomy. Conventional political theory can narrate this history but cannot predict it. Why does a technically superior constitutional arrangement (compliance with foundational law) lose systematically to an irregular one? The answer requires evolutionary analysis.

Darwin provided the missing conceptual framework, but political science never integrated it rigorously. Evolutionary thinking occasionally surfaces in political analysis, talk of "political survival," "fitness for office," or "adaptation to circumstances", but these remain metaphors rather than rigorous theoretical imports. The core Darwinian insight is that selection operates on differential reproduction, not conscious intention or functional superiority. Organisms evolve traits that maximize genetic replication, which may or may not align with organismic welfare. The peacock's tail is extravagant, energy-intensive, and increases predation risk, yet it persists because it maximizes reproductive success through sexual selection. The tail succeeds not by making peacocks healthier or longer-lived but by making them more likely to reproduce. Similarly, institutions evolve features that maximize memetic replication, which may or may not align with social welfare. The question is never "does this institution work well?" but rather "does this institutional configuration replicate more successfully than alternatives?"

Richard Dawkins's The Extended Phenotype (1982) extended Darwin's logic in a direction directly applicable to political institutions. Dawkins argued that phenotypic effects of genes, the physical expressions of genetic information, are not limited to the organism's body. A beaver's genes express themselves not only in the beaver's fur, teeth, and physiology but also in the beaver dam the animal constructs. The dam is as much a phenotypic expression of beaver genes as the beaver's tail. Crucially, the dam persists after the individual beaver dies, continues to shape the environment, and affects the survival probability of the genes that built it. The dam is extended phenotype, environmental modification that extends the gene's causal influence beyond the organism's skin.

Legal institutions are extended phenotypes of memetic replicators. A constitution is not merely a "representation" of political ideas but a material artifact, paper, ink, institutional infrastructure: that persists beyond its creators and shapes the environment in ways that affect its own replication probability. Argentina's labor law regime, crystallized in constitutional Article 14bis (1957) and elaborated through judicial doctrine and administrative practice, functions identically to a beaver dam. It persists across administrations of wildly different ideological orientation, military dictatorships, Peronist governments, neoliberal reformers, and libertarian revolutionaries, because it has constructed an institutional ecology that makes its own dismantling prohibitively costly. The labor regime is not a "reflection" of Argentine political culture but phenotypic machinery that actively shapes culture in ways that ensure its own survival.

Daniel Dennett's Darwin's Dangerous Idea (1995) clarified the epistemological implications. Darwinian reasoning is what Dennett called "universal acid": it dissolves essentialist thinking wherever applied. If we accept that cultural evolution proceeds through variation, selection, and differential replication of memes, units of cultural information, then we must abandon the assumption that cultural institutions exist because they serve collective functions. The functionalist assumption pervades political theory: institutions exist because they solve coordination problems, enforce agreements, or aggregate preferences. Evolutionary analysis inverts this logic. Institutions exist because they successfully replicate, which may coincidentally serve collective functions but need not. The relevant question is not "what function does this institution serve?" but "what features enable this institutional configuration to outcompete alternatives in the memetic environment?"

This paper synthesizes Machiavellian political realism with Darwinian evolutionary epistemology. Machiavelli provided the method, describe power as it operates, not as it should operate. Darwin, Dawkins, and Dennett provide the framework, institutions are replicators subject to selection pressures, optimizing for fitness rather than welfare, constructing extended phenotypes that alter their own selection environment. The synthesis generates a research program focused on institutional evolutionary dynamics: measuring memetic fitness, predicting transplant success, quantifying lock-in, and mapping fitness landscapes.

The paper proceeds in four movements. Section II establishes the core principle: institutions optimize for memetic fitness, not social welfare. This fitness-welfare divergence explains the persistence of manifestly dysfunctional arrangements. Section III applies Evolutionarily Stable Strategy analysis to institutional competition, explaining why superior alternatives fail to displace inferior incumbents. Section IV develops the extended phenotype framework systematically, showing how three major anomalies that puzzle conventional theory, transcultural legal convergence, systematic transplant failure, and the WEIRD implementation gap, follow naturally from memetic selection. Section V examines the ultraactivity trap, a particularly destructive form of institutional lock-in that transforms repeated democratic games into terminal competition. Section VI concludes by distinguishing evolutionary analysis from naturalistic fallacy and sketching implications for 21st-century governance.

The core thesis can be stated starkly: The modern governor who understands evolutionary biology has decisive advantage over one relying on conventional political theory. Not because evolution justifies governance: that would be naturalistic fallacy, but because evolutionary analysis enables prediction of which reforms persist and which collapse, independent of technical merit or moral worth. Where Machiavelli

taught princes to acquire and maintain personal authority through cunning adaptation to circumstance, evolutionary analysis teaches governors to engineer institutional ecosystems that replicate themselves across generations. The difference is profound. Personal authority dies with the prince. Institutional ecosystems, once established, survive indefinitely, or until a sufficiently severe environmental shock disrupts the equilibrium they constructed.

## II. FITNESS VS. WELFARE: THE CORE PRINCIPLE

The central principle of evolutionary institutional analysis can be stated with deceptive simplicity: institutions optimize for replication, not results. This fitness-welfare divergence explains patterns that appear paradoxical or irrational under conventional analysis but follow naturally from selection logic. An institution that maximizes social welfare but replicates poorly will be outcompeted by an institution that replicates reliably despite producing suboptimal outcomes. Selection operates on what reproduces, not on what functions well. This principle is not normative but descriptive: not a claim about how institutions should work but an observation about how they actually work under selection pressure.

The biological analogy clarifies the logic. Consider the peacock's tail, Darwin's own example of sexual selection. The tail is metabolically expensive, requiring substantial energy to grow and maintain. It impairs flight, making peacocks more vulnerable to predation. It serves no obvious survival function, indeed, it actively harms individual peacock welfare by increasing mortality risk. Yet tails persist and even elaborate across generations because they enhance reproductive success. Peahens preferentially mate with males displaying more extravagant tails, possibly because tail quality honestly signals male genetic fitness, only healthy males can afford the metabolic cost. The tail succeeds not by promoting peacock welfare but by promoting peacock gene replication. The trait persists because bearers reproduce at higher rates, not because they survive longer or live better.

Institutional persistence follows identical logic. Consider Argentine labor law, encoded in constitutional Article 14bis (1957) and elaborated through six decades of judicial doctrine and administrative practice. The regime mandates extensive protections: employment stability, union representation, profit-sharing, social security, comprehensive workplace regulation. Economically, these mandates increase unemployment (by raising formal sector labor costs), reduce productivity (through rigid employment rules), and distort capital allocation (through mandatory profit-sharing).

Argentine informal sector employment fluctuates between 25-35% of the workforce, substantially higher than comparable Latin American countries. Productivity growth has averaged 0.4% annually since 1960, compared to 2.1% for Chile and 1.8% for Uruguay. Capital investment as percentage of GDP has declined from 25% (1960-1974) to 16% (2000-2024). By conventional welfare metrics, the labor regime produces suboptimal outcomes.

Yet the regime persists with extraordinary stability. It survived military dictatorships (1966-1973, 1976-1983) that explicitly targeted it for elimination. It survived hyperinflation (1989-1990) that collapsed the currency and much institutional infrastructure. It survived the Washington Consensus reforms (1990s) that dismantled state enterprises and deregulated multiple sectors. It survived default and economic implosion (2001-2002) that toppled governments and triggered institutional crisis. Most tellingly, it survived repeated direct assault by governments elected on explicit mandates to reform it. The Macri administration (2015-2019) dedicated substantial political capital to labor reform, hiring international consultants, drafting careful legislation, cultivating business support. The reforms passed Congress with difficulty in 2017. By 2023, virtually all substantive changes had been reversed through judicial interpretation, administrative practice, or direct legislative repeal. The labor regime exhibits what I have elsewhere termed Constitutional Lock-in Index (CLI) of 0.89, near-maximum institutional crystallization on a 0-1 scale where 1 represents complete rigidity.

Why does a welfare-reducing institution outcompete welfare-enhancing alternatives? The answer lies in differential replication, not functional superiority. The labor regime replicates through several mechanisms that have nothing to do with economic outcomes. First, it creates a constituency, unionized formal sector workers, whose material interests align with regime preservation. This constituency comprises approximately 30% of the electorate but punches above its electoral weight through organizational capacity and strategic voting concentration in urban districts. Any government contemplating labor reform must calculate whether potential efficiency gains exceed certain electoral costs. Since efficiency gains are diffuse, long-term, and uncertain while electoral costs are concentrated, immediate, and certain, the political calculus favors preservation.

Second, the regime embeds itself in interpretive infrastructure. Six decades of judicial precedent, administrative regulation, and professional practice have built an elaborate apparatus for interpreting and applying labor law. This apparatus employs tens of thousands of people, judges, clerks, labor inspectors, arbitrators, union officials, management lawyers, who have developed expertise specific to existing

arrangements. Reform threatens specialized human capital, triggering organized resistance from professional groups whose livelihoods depend on current complexity. More subtly, the apparatus shapes how labor issues are cognitively framed. Labor conflicts are automatically interpreted through concepts like “acquired rights,” “regulatory degradation,” and “worker vulnerability” embedded in existing doctrine. Alternative framings, “employment barriers,” “productivity constraints,” “investor uncertainty”, struggle for cognitive traction because they lack equivalent institutional embodiment.

Third, the regime has constructed what I term immunological defenses, interpretive mechanisms that transform criticism into confirmation. Labor reform advocacy is interpreted as “neoliberal attack on worker rights,” which activates pre-existing narratives about oligarchic resistance to popular sovereignty. Economic dysfunction attributable to labor rigidity is reinterpreted as “capitalist sabotage” demonstrating the necessity of stronger protections. Even crisis that temporarily weakens the regime, hyperinflation, default, unemployment, ultimately strengthens it by validating the narrative that workers require protection against market volatility. The regime has achieved what Karl Popper warned against in scientific theories but what succeeds remarkably well for ideological systems: it has made itself unfalsifiable. Any evidence can be interpreted as confirming regime necessity.

These replication mechanisms are entirely independent of economic performance. Indeed, economic dysfunction may even enhance memetic fitness by intensifying the perceived need for protection. A labor regime in a thriving economy might appear dispensable; a labor regime in a chronically dysfunctional economy appears indispensable. The regime replicates not despite producing poor outcomes but partly because poor outcomes validate its existence. This is fitness-welfare divergence in pure form.

The principle generalizes beyond labor law to institutional analysis broadly. Any institution optimizing for replication will evolve features that enhance survival and transmission probability, which may or may not align with ostensible institutional purposes. Regulatory agencies evolve to maximize agency survival (budgets, authority, autonomy) rather than regulatory effectiveness. Judicial doctrines evolve to maximize doctrine persistence (citing precedent, resisting reversal, building protective complexity) rather than legal clarity or justice. Constitutional arrangements evolve to maximize arrangement stability (raising amendment barriers, concentrating interpretive authority, cultivating defending coalitions) rather than adaptive governance or welfare maximization.

This is not a claim that institutions never serve collective functions. Rather, when institutional features serve collective welfare, this is typically because welfare-enhancement coincidentally promotes replication, not because welfare is the optimization target. Institutions that harm their hosts eventually disappear through host collapse, imposing selection pressure for some minimal functionality. But this constraint is loose, many institutions persist for decades or centuries while producing poor outcomes, eventually collapsing but only after extraordinary longevity relative to human lifespans or even national histories. Argentine labor law has persisted 80 years despite chronic economic underperformance. Soviet central planning persisted 70 years despite productivity stagnation. The Habsburg Empire persisted 400 years despite repeated military defeat and administrative dysfunction. From an evolutionary perspective, these are success stories, institutions that successfully replicated across many generations, even if from a welfare perspective they represent failure.

The fitness-welfare divergence is most starkly visible in legal transplants, taking a law that functions well in one jurisdiction and implementing it in another. Argentina's Law 27,401 (2017) provides a paradigmatic case. The law established corporate compliance regimes for corruption prevention, closely modeled on successful Spanish (Organic Law 1/2015) and Chilean (Law 20,393) precedents. Technical quality was high, international consultants drafted the law, incorporating best practices from OECD guidelines and comparative analysis of successful regimes. Enforcement has been robust, the Anti-Corruption Office (Oficina Anticorrupción) conducted regular inspections, the judiciary has prosecuted violations, and administrative sanctions have been consistently applied. Economic logic is compelling, corruption costs Argentina an estimated 5% of GDP annually, and compliance regimes have demonstrated effectiveness in reducing corruption in comparable jurisdictions.

Yet adoption has been catastrophically low. As of 2024, only 3.8% of Argentine companies subject to the law have implemented meaningful compliance programs. This compares to 67% adoption in Spain and 71% in Chile, despite Argentina's theoretically superior enforcement. The pattern is precisely backwards from what rational choice predicts: stronger enforcement produces lower compliance. My analysis using the IusMorfos framework, which measures environmental compatibility across cultural, institutional, path dependency, economic, and technical dimensions, assigned Law 27,401 an Environmental Compatibility score of 0.31, well below the 0.65 threshold I have found predicts transplant success. The prediction was failure; the observation confirmed it.

The divergence between technical quality (high) and memetic fitness (low) illuminates the mechanism. Spanish and Chilean compliance regimes evolved within what I term

“cooperative compliance” cultures, where regulatory relationships are understood as partnerships between regulators and regulated entities for achieving shared objectives. The regulator provides guidance and technical assistance; the regulated entity implements programs in good faith; enforcement is reserved for bad actors who reject cooperation. This cultural logic is embodied in regulatory practice, professional training, judicial interpretation, and business norms. It is not written in the law, indeed, laws in all three countries use similar mandatory language, but it pervades the institutional ecosystem.

Argentine regulatory culture operates on fundamentally different principles. Regulatory relationships are understood as adversarial, regulators seek to extract penalties; regulated entities seek to minimize compliance; both assume the other acts in bad faith. This adversarial logic is equally embedded in institutional practice, professional training, and interpretive conventions. When Law 27,401 was transplanted, it imported the Spanish/Chilean legal text but not the cooperative culture making that text functional. Argentine companies interpreted the compliance mandate as another opportunity for regulatory predation. Regulators interpreted non-compliance as confirmation of bad faith requiring aggressive enforcement. The result was a self-fulfilling equilibrium where aggressive enforcement triggered defensive non-compliance, which justified more aggressive enforcement, and so forth.

The law failed not because it was technically deficient: it was technically superior, but because it had low memetic fitness in Argentine institutional ecology. A compliance regime adapted to cooperative cultures has traits that actively harm reproduction in adversarial cultures. Voluntary disclosure, for example, is adaptive in cooperative environments (builds trust, facilitates assistance) but maladaptive in adversarial environments (provides ammunition for prosecution). Detailed documentation is adaptive cooperatively (demonstrates good faith, supports continuous improvement) but maladaptive adversarially (creates liability evidence, increases inspection surface). The transplanted meme carried traits that made sense in its native ecology but proved toxic when transplanted.

This pattern is not unique to Argentina or to compliance regimes. My broader analysis of legal transplants across 60 cases in Latin America (2000-2024) found 76% failure rates, failure defined as substantial divergence from source jurisdiction performance. Strikingly, failure rates showed no correlation with technical quality ($r=0.12$, statistically insignificant) but strong correlation with environmental compatibility ($r=0.87$, $p<0.001$). The IusMorfos framework, which systematically measures compatibility across five dimensions weighted by their predictive power, achieves 87.4% accuracy in predicting transplant outcomes. Technical assessment alone

achieves 61% accuracy, barely better than chance for binary predictions. The difference is that technical assessment evaluates law as text while environmental compatibility evaluates law as meme embedded in ecosystem.

The fitness-welfare divergence has profound implications for reform strategy. Conventional reform logic assumes that demonstrating dysfunction creates pressure for improvement, show leaders that current arrangements produce poor outcomes, propose technically superior alternatives, expect rational adoption. This logic fails systematically because it misunderstands the optimization target. Institutions are not optimizing for outcomes but for replication. Demonstrating that an institution produces poor outcomes does not create pressure for replacement if the institution replicates reliably despite poor outcomes. Indeed, as noted, poor outcomes may strengthen replication by intensifying perceived need for institutional protections.

Successful reform must instead target replication mechanisms directly. This requires analyzing: (1) which constituencies depend materially on current arrangements and could be offered alternative dependencies; (2) what interpretive infrastructure embeds the institution and could be redirected through reframing; (3) what immunological defenses protect the institution and could be circumvented through approach variation. None of these questions concern institutional performance. All concern institutional reproduction.

The strategic implication is counterintuitive but follows rigorously from evolutionary logic. A reform has higher probability of success if it offers superior memetic fitness than if it offers superior welfare outcomes. This does not mean reformers should ignore welfare, welfare improvement may coincidentally enhance fitness, and demonstrating welfare benefits may assist in building supporting coalitions. But fitness enhancement must be the primary design criterion. A reform optimized for welfare but neglecting fitness will fail; a reform optimized for fitness but producing modest welfare improvements may succeed. The modern prince learns: design for replication, not results. If results matter, ensure they enhance replication.

## III. EVOLUTIONARILY STABLE STRATEGIES IN GOVERNANCE

Machiavelli knew that conquering a state is easier than holding it. In Il Principe, he devoted substantial analysis to acquisition strategies, inheritance, fortune, arms, criminal cunning, but acknowledged that acquisition means little without retention. He attributed retention failure to fortuna (external circumstance) and human fickleness, which are descriptions rather than explanations. Evolutionary game theory provides the mechanism Machiavelli lacked: Evolutionarily Stable Strategies (ESS). An ESS is a

strategy that, once established in a population, resists invasion by alternative strategies. The concept, developed by John Maynard Smith for analyzing animal behavior, applies with equal force to institutional competition. The critical insight is that ESS can be locally stable but globally suboptimal, inferior strategies can persist indefinitely if they have achieved evolutionary stability, even when superior alternatives exist.

The formal definition of ESS involves game-theoretic payoffs, but the intuition is straightforward. A strategy S is evolutionarily stable if it satisfies two conditions: first, when the entire population plays S, no individual can gain advantage by unilaterally deviating to strategy S'; second, if S and S' are equally successful when S dominates the population, then S must do better than S' when the population contains many S' players. The first condition is equilibrium: no one benefits from unilateral deviation. The second condition is stability: if an alternative strategy invades the population at low frequency (through mutation, immigration, or innovation), it cannot increase in frequency even if it performs equally well against the dominant strategy. The alternative must actively outperform the incumbent when rare, not merely match its performance. This is an extremely demanding requirement, which is precisely why ESS are stable.

The relevance to institutional persistence is immediate. Institutional arrangements can be understood as strategies in repeated games involving multiple actors, politicians, bureaucrats, judges, businesses, voters, unions, civil society. These actors operate within inherited institutional frameworks that structure payoffs, define permissible actions, and shape expectations about others' behavior. Once an institutional arrangement achieves ESS status, attempts to reform it face structural disadvantage. A reform must not merely offer better outcomes but must do so when operating within a population still dominated by actors adapted to the incumbent arrangement. This is frequently impossible. The incumbent strategy succeeds because it has shaped the environment, material incentives, cognitive categories, professional expertise, organizational forms, to favor its own reproduction. Alternative strategies designed for different environments find themselves operating in hostile ecology.

Argentine populism provides a paradigmatic case of institutional ESS. The memetic complex I term "populist strategy", combining nationalist rhetoric, labor protections, social welfare programs, discretionary fiscal policy, and adversarial relationships with markets and institutions, was established by the first Perón administration (1946-1955). It has persisted with remarkable consistency for 80 years despite repeated attempts at displacement. The term "populism" is analytically contested and often deployed pejoratively, but I use it descriptively to denote a specific bundle of

institutional arrangements and policy commitments that have demonstrated multi-generational reproductive success in Argentine political ecology.

The historical record documents repeated failed attempts to establish alternative strategies. The Revolución Libertadora military government (1955-1958) explicitly targeted Peronist institutional infrastructure for elimination, dissolving the Peronist party, purging the bureaucracy, and attempting to de-Peronize unions. The regime failed. Perón returned to power in 1973, and many Peronist structures reconstituted themselves. The Onganía dictatorship (1966-1973) attempted technocratic state-led development as alternative to Peronist populism. It lasted seven years before collapsing amid social upheaval, returning power to Peronism. The Proceso dictatorship (1976-1983) waged a far more brutal campaign of institutional restructuring, combining violent repression with orthodox economic liberalization. After seven years and an estimated 30,000 deaths, the regime collapsed in military defeat (Malvinas War) and economic crisis, again returning power to Peronism.

The democratic period (1983-present) shows an equally striking pattern. Argentina has experienced three governments elected on explicitly anti-Peronist, economically liberal platforms: Alfonsín's Radical government (1983-1989), De la Rúa's Alliance government (1999-2001), and Macri's Cambiemos government (2015-2019). None completed a full eight-to-ten-year political cycle, the period I have identified as necessary for institutional reforms to achieve reproductive maturity. Alfonsín lasted six years before hyperinflation forced early resignation. De la Rúa lasted two years before economic collapse and political crisis forced his departure via helicopter from the presidential palace. Macri completed his constitutional term but his reform program was substantially reversed during his own tenure through judicial interpretation and administrative resistance, with the remainder reversed by his Peronist successor (2019-2023).

The pattern is not random. The consistent failure of liberal governments to complete reproductive cycles while Peronist governments regularly do. Menem (1989-1999), Kirchner (2003-2007), Fernández de Kirchner (2007-2015), Fernández (2019-2023), suggests systematic rather than contingent causation. The explanation lies in differential fitness. Populist strategy has several features that enhance reproduction in Argentine institutional ecology, features that liberal strategy lacks.

First, populist strategy exhibits high cognitive simplicity. Political messaging reduces to binary oppositions: pueblo versus oligarchy, nation versus colonialism, majority versus privilege. These dichotomies are easily understood, emotionally resonant, and cognitively "sticky": they persist in memory and frame subsequent perception. Liberal

strategy messaging involves complex causal chains: deregulation enables competition which drives productivity which increases wages which generates sustainable growth. The causal chain is empirically accurate but cognitively costly. It requires holding multiple steps in memory, understanding temporal sequencing, and resisting immediate intuition (deregulation can appear to favor business over workers, even if ultimate effect is opposite). In memetic competition, simple messages outcompete complex messages if both achieve similar levels of initial exposure.

Second, populist strategy offers high temporal gratification. Redistributive programs provide immediate benefits to large constituencies, pensioners receive higher benefits, workers gain job security, consumers see subsidized prices. Costs are deferred, diffuse, or externalized; inflation erodes benefits gradually, fiscal deficits accumulate slowly, productivity stagnation manifests over decades. Liberal strategy inverts this temporal structure. Reforms impose immediate costs (fiscal adjustment reduces transfers, labor flexibility increases unemployment risk, price deregulation raises consumer costs) while promising delayed, diffuse benefits (higher growth, lower inflation, better employment). Memetic fitness strongly favors immediate gratification over delayed benefits because political time horizons are short. Voters evaluate based on recent experience; delayed benefits may accrue after electoral accountability has shifted to successors.

Third, populist strategy enjoys high cultural compatibility with Argentine political culture, which has been shaped by 80 years of populist hegemony. Populist concepts: social rights, state responsibility, market suspicion are deeply embedded in judicial doctrine, administrative practice, media discourse, educational curriculum, and popular consciousness. Liberal concepts -market efficiency, limited government, individual responsibility- are present but marginal, frequently framed as foreign impositions or elite interests. This cultural embedding is both cause and consequence of populist persistence, creating self-reinforcing dynamics. Populist strategy shapes culture in ways that favor populist reproduction; culture shaped by populism disadvantages liberal alternatives.

These fitness advantages compound multiplicatively. A strategy that scores high on simplicity, gratification, and compatibility has dramatically higher reproductive success than a strategy scoring low on all three dimensions. My analysis of Argentine political cycles (1946-2025) identifies what I term a memetic fitness ratio of approximately 216:1; populist memes reproduce 216 times more successfully than liberal memes in Argentine ecology. This is not a precise measurement, but an order-of-magnitude estimate derived from comparing tenure lengths, reform persistence rates, and electoral success patterns. The key insight is not the specific number but

the mechanism: populist strategy has achieved ESS status through fitness advantages that have nothing to do with outcome quality.

The ESS interpretation explains several otherwise puzzling patterns. Why did the Menem administration (1989-1999), despite implementing orthodox economic liberalization -privatization, trade opening, fiscal discipline- remain categorized as Peronist rather than liberal? Because Menem preserved the symbolic and institutional core of populist strategy (nationalist rhetoric, union relationships, discretionary fiscal capacity) even while pursuing liberalizing reforms. The Menem strategy was parasitic on populist ESS -using populist legitimacy to implement liberal policies- rather than attempting to establish a competing liberal ESS. When Menem's successor De la Rúa attempted pure liberal strategy without populist legitimation, the strategy collapsed within two years. This confirms that liberalization succeeded when embedded within populist ESS but failed when attempted as standalone alternative.

Why did the Macri administration, despite explicit anti-Kirchnerist mandate and strong initial support, fail to consolidate institutional reforms? Because Macri attempted gradual reform -"cambio gradual"- which is precisely wrong from ESS perspective. Gradual reform allows the incumbent ESS to adapt, mobilize resistance, and reconstitute itself. It provides time for the immunological defenses I discussed in Section II to activate. From ESS analysis, successful displacement requires rapid, comprehensive change that doesn't give the incumbent strategy time to respond. This was the New Zealand lesson (1984-1993), which I will examine shortly. Macri's gradual strategy allowed Peronist opposition to organize, frame reforms as elite impositions, cultivate judicial resistance, and ultimately recapture power with mandate to reverse reforms.

Why do Argentine economic crises -which have been frequent and severe (1975, 1981-1982, 1989-1990, 2001-2002, 2018-2019)- consistently lead to return of populist government rather than liberal reform? Conventional wisdom holds that crisis creates opportunity for reform because status quo legitimacy collapses. This is half right. Crisis does collapse status quo legitimacy, but that doesn't automatically create opening for liberal alternatives. Rather, crisis activates populist interpretive frameworks -market failure, foreign conspiracy, elite betrayal- that are already embedded in institutional infrastructure and popular consciousness. Liberal interpretations -government failure, distorted incentives, populist excesses- must compete against pre-existing populist frameworks from position of disadvantage. The crisis empowers whichever strategy has better embedded itself in interpretive infrastructure, which is typically the incumbent ESS.

The New Zealand case provides crucial contrast. In 1984, New Zealand faced severe economic crisis: foreign exchange shortage, near-default, capital controls, wage-price controls, triple-digit inflation. The newly elected Labour government, traditionally aligned with statist policies, implemented radical liberalization over 18-24 months: floated currency, deregulated interest rates, removed capital controls, eliminated subsidies, reformed tax system, corporatized state enterprises. The reforms persisted across four subsequent governments (National, Labour, National, Labour again) and remain largely intact four decades later. New Zealand achieved what Argentina has attempted six times and failed six times: genuine ESS transition.

The New Zealand case illuminates the conditions for displacing incumbent ESS. I identify four factors that were present in New Zealand but absent in Argentine reform attempts: crisis severity, coalition breadth, implementation speed, and institutional irreversibility. First, the New Zealand crisis was externally caused and universally acknowledged. The foreign exchange constraint was technical reality, not subject to interpretive dispute. Domestically-caused crises can be reinterpreted through existing frameworks ("capitalist sabotage," "neoliberal policies"), but external constraints impose themselves regardless of interpretation. Second, Labour's implementation of liberal reforms created broad coalition spanning traditional political divides. Business supported reforms; unions grudgingly accepted them because implemented by "their" party; middle-class voters saw no plausible alternative. Argentine reforms were consistently implemented by right-of-center governments opposed by left-of-center Peronism, creating clear partisan divides rather than cross-partisan acceptance.

Third, New Zealand's implementation speed was rapid: 18-24 months for major structural changes. This speed prevented effective opposition mobilization. By the time opponents organized, reforms had already generated constituencies with vested interests in maintenance. Argentine reforms were gradual: Macri planned four-year implementation schedules, which provided ample time for opposition coordination. Fourth, New Zealand embedded reforms in institutional structure that made reversal costly. The Reserve Bank Act (1989) gave the central bank statutory inflation-targeting mandate with governor dismissible for failure. This created credibility with markets and constituents that would be destroyed by reversal. Argentine reforms remained administrative -presidential decrees, ministerial directives- easily reversed by successors without institutional cost.

The contrast reveals that ESS displacement is possible but requires extraordinary conditions. The incumbent ESS has structural advantages -embeddedness in institutions, culture, cognitive frameworks- that challengers lack. Successful displacement requires simultaneously achieving crisis that discredits incumbent,

coalition that spans traditional divides, speed that prevents adaptation, and irreversibility that raises reversal costs. Achieving any single condition is difficult; achieving all four simultaneously is rare. This explains why ESS persist despite inferior performance. The performance inferiority may be substantial, but the structural advantages of incumbency mean that displacement requires near-perfect execution of reform strategy.

The ESS framework generates testable predictions. The current Milei administration (2023-present) faces the classic reformer's challenge: displacing populist ESS with liberal alternative. We can assess ex ante whether conditions favor success. Crisis severity: high: Argentina faces 211% inflation, negative international reserves, 41% poverty rate. Coalition breadth: low: Milei governs with minority in both legislative chambers, opposition controls key governorships, unions are hostile. Implementation speed: constrained, the Ley Omnibus, Milei's comprehensive reform package, was substantially diluted in Congress; implementation is proceeding through presidential decree (DNU), which is constitutionally fragile. Institutional irreversibility: weak: reforms are administrative rather than constitutional; no independent institutions have been created to lock in changes; reversal by successor would be legally straightforward.

The prediction from ESS analysis is conditional but clear. If Milei achieves rapid implementation before opposition consolidates and embeds reforms in irreversible institutional structure, there is perhaps 20-30% probability of successful ESS displacement. If implementation remains gradual and reversible, probability is near zero. The relevant comparison is not between Milei's policies and Peronist policies on economic merits, his policies may well be superior on conventional metrics, but rather between the memetic fitness of liberal strategy in Argentine ecology (historically low) versus the embedded advantages of populist ESS (historically overwhelming). Past base rate is 0 of 6 liberal governments successfully completing reproductive cycles. Milei would need to be historically exceptional to overcome these odds.

The ESS framework transforms institutional analysis from outcome evaluation to reproduction analysis. The relevant question is never "does this institutional arrangement work well?" but rather "does this arrangement outcompete alternatives in replication?" An ESS that produces chronically inferior outcomes but reliably outreproduces challengers will persist indefinitely, or until environmental shock sufficiently severe to destabilize the equilibrium. The modern prince learns: established strategies resist displacement even when manifestly inferior. Successful reform requires not just better policy but better memetic fitness, achieved through crisis exploitation, coalition construction, rapid execution, and institutional

entrenchment. Three of four guarantees failure; achieving all four simultaneously is historically rare and demands political acumen of the highest order.

## IV. LAW AS EXTENDED PHENOTYPE: RESOLVING THREE ANOMALIES

Richard Dawkins's The Extended Phenotype (1982) argued that gene effects extend beyond organism bodies. Beaver genes express themselves not only in beaver physiology, teeth, tail, fur, but also in the beaver dam. The dam is phenotypic expression as much as the beaver's tail. Crucially, the dam persists after individual beavers die, continues shaping the environment, and affects survival probability of the genes that built it. The dam is extended phenotype, environmental modification that extends gene causal influence beyond organism boundaries. Legal institutions are extended phenotypes of memetic replicators. This section demonstrates that three major anomalies puzzling conventional theory, transcultural convergence, systematic transplant failure, and the WEIRD implementation gap, follow naturally from extended phenotype analysis.

**Anomaly One: Transcultural Normative Convergence Without Social Sources**

George Murdock's Ethnographic Atlas documents marriage and kinship patterns across 1,267 human societies spanning all continents, linguistic families, and subsistence modes. The dataset includes societies with no historical contact, radically different economic systems, incompatible cosmologies, and antithetical social structures. Yet across this extraordinary diversity, one pattern exhibits near-universal convergence: incest prohibition. Of 1,267 societies, 1,257 prohibit sexual relations between parents and children or between siblings, 99.2% convergence. When narrowed to parent-child and sibling prohibitions specifically, convergence reaches 94.7%. This is not cultural diffusion, many societies had zero contact with

others exhibiting identical prohibitions. It is not functional necessity in any obvious sense, incest prohibition does not solve coordination problems or enforce agreements. It is not divine command, societies converge on prohibition whether they believe in zero gods, one god, or many gods.

Legal positivism cannot explain this convergence. H.L.A. Hart's rule of recognition grounds law validity in social practice within bounded legal systems. There is no transsocietal recognition rule spanning disconnected societies. Convergence on incest prohibition occurs absent the social sources positivism requires for legal

validity. Natural law theory might seem to explain the pattern, perhaps incest prohibition reflects natural moral law apprehensible by human reason. But natural law predicts convergence on all fundamental moral truths, not selective convergence. Why near-universal convergence on incest but dramatic divergence on property rights, political authority, distributive justice, and punishment? If human reason naturally apprehends moral truth, why does it apprehend incest prohibition universally but property arrangements not at all?

Extended Phenotype Theory resolves the anomaly straightforwardly. Incest prohibition demonstrates fitness advantage at population level independent of conscious understanding. Populations prohibiting close genetic inbreeding experience higher offspring viability through reduced expression of recessive deleterious alleles and increased immune system diversity. This fitness advantage operates through demographic selection across generations, populations practicing systematic inbreeding decline relative to outbreeding populations, not because anyone understands genetics but because outbreeding populations simply reproduce more successfully. Cultural evolution selects for norms that enhance group reproductive success, which may or may not align with conscious intentions or stated justifications.

The mechanism requires no intentional design. Societies develop various marriage rules and kinship prohibitions through cultural innovation and variation. Those prohibitions that coincidentally reduce close inbreeding lead to population expansion through higher offspring survival and fertility. Societies lacking such prohibitions experience relative demographic decline. Over sufficient timescales, hundreds of generations, selection produces convergence on approximately optimal inbreeding avoidance without anyone understanding the mechanism. The prohibition persists not because it is morally correct or functionally necessary but because societies practicing it outreproduce societies lacking it.

Inbreeding depression coefficient estimates suggest first-cousin marriage reduces offspring fitness by approximately 27% through increased infant mortality, childhood illness, and reduced fertility. Closer genetic relationships, siblings, parent-child, produce far more severe fitness penalties. These fitness effects compound across generations. A society practicing systematic close inbreeding will be demographically overwhelmed within 10-15 generations by neighboring societies avoiding inbreeding, regardless of military prowess, technological sophistication, or economic productivity. Selection at population level is brutal and effective.

The transcultural convergence thus demonstrates extended phenotype logic. Incest prohibition is phenotypic machinery constructed by memes encoding kinship norms.

The machinery persists because it enhances reproductive success of the populations bearing it. The machinery does not require anyone to understand why it works, indeed, most societies justify incest prohibition through moral disgust, supernatural contamination, or social order rather than genetics. The stated justifications are post hoc rationalizations, not causal mechanisms. The actual cause is differential reproduction at population level, which is invisible to participants but decisive for outcomes.

**Anomaly Two: Systematic Legal Transplant Failure Despite Technical Equivalence**

Legal transplants, adopting law from successful source jurisdiction and implementing it in recipient jurisdiction, fail at 76% rates globally. This failure rate is particularly puzzling when transplants copy technically sophisticated laws from well-functioning legal systems into jurisdictions sharing similar legal traditions, economic development, and institutional capacity. My analysis of 60 transplant cases in Latin America (2000-2024) documented pervasive failure despite apparent similarity between source and recipient jurisdictions.

The Law 27,401 case illuminates the mechanism. Argentina transplanted corporate compliance law from Spain and Chile, countries sharing Spanish legal tradition, similar economic development, and comparable corruption problems. The transplanted law was technically excellent, carefully drafted by international consultants incorporating OECD guidelines and comparative best practices. Enforcement was strong, the Anti-Corruption Office conducted systematic inspections, courts prosecuted violations, administrative penalties were consistently imposed. Yet adoption reached only 3.8%, compared to 67% in Spain and 71% in Chile.

Conventional theories fail to explain this outcome. Rational choice predicts that technically superior law with strong enforcement should produce high compliance. Historical institutionalism might invoke path dependency but cannot explain why dependency blocks apparently minor adaptations. Cultural theory might point to general anti-compliance attitudes, but Argentina achieves normal compliance rates in other regulatory domains, tax filing, vehicle registration, construction permits. The failure is specific to this transplant, not general to regulation.

Extended Phenotype Theory explains the failure through environmental compatibility. Law 27,401 is a meme adapted to “cooperative compliance” ecology, where regulatory relationships are understood as partnerships between regulators and regulated entities pursuing shared objectives. This cultural understanding is embedded in Spanish and Chilean regulatory practice, professional training, judicial interpretation, and business norms. It shapes how compliance requirements are communicated, how

implementation problems are resolved, and how good-faith efforts are distinguished from bad-faith evasion. The legal text itself is content-neutral on these points: it mandates compliance without specifying relationship quality. But compliance success depends critically on these unwritten cultural features.

Argentine regulatory ecology operates on fundamentally different principles. Regulatory relationships are understood as adversarial, regulators seek to extract penalties, regulated entities seek to minimize compliance, both assume the other operates in bad faith. This adversarial logic is equally embedded in institutional practice, professional culture, and behavioral norms. When Law 27,401 was transplanted, it imported the Spanish/Chilean legal text but not the cooperative ecology making that text function. The text, optimized for cooperative environment, contains features actively maladaptive in adversarial environment.

Consider voluntary disclosure requirements. In cooperative ecology, voluntary disclosure builds trust, facilitates regulator assistance, and reduces monitoring costs. Companies disclose problems because they expect collaborative problem-solving. In adversarial ecology, voluntary disclosure provides ammunition for prosecution. Companies interpreting disclosure requirements through adversarial lens naturally minimize disclosure, triggering regulator suspicion that justifies aggressive inspection, which confirms company fears, producing self-reinforcing adversarial equilibrium. The law fails not because companies are irrational but because rational response to the law depends on ecological context absent from legal text.

I have developed a computational framework. IusMorfos, for predicting transplant success through environmental compatibility assessment across five dimensions: cultural compatibility (35% weight), institutional ecology (25%), path dependency (20%), economic context (15%), and technical quality (5%). The weighting reflects predictive power derived from empirical calibration. For Law 27,401, the assessment yields Environmental Compatibility score of 0.31, well below the 0.65 threshold I have found distinguishes success from failure. The score breaks down as follows: cultural compatibility 0.22 (cooperative versus adversarial regulatory philosophy), institutional ecology 0.28 (weak horizontal enforcement despite strong vertical enforcement), path dependency 0.41 (labor law ultraactivity shapes corporate law interpretation), economic context 0.35 (informal economy 25% GDP), technical quality 0.92 (law is technically excellent).

The low overall compatibility score predicted failure. The observed 3.8% adoption confirmed the prediction. Crucially, technical quality contributes only 5% to overall compatibility because technical assessment evaluates law as text while compatibility

evaluates law as meme embedded in ecosystem. A technically perfect law incompatible with host ecology will fail; a technically imperfect law compatible with host ecology will succeed. This inverts conventional reform logic, which prioritizes technical quality and assumes implementation follows automatically from sound legal design.

The IusMorfos framework achieves 87.4% accuracy across 60 transplant cases, substantially outperforming technical assessment (61% accuracy) or moral merit assessment (43% accuracy). The superior performance confirms that transplant success depends primarily on environmental compatibility rather than intrinsic legal quality. This has direct implications for reform strategy. Before transplanting law, reformers must assess ecosystem compatibility. If compatibility is low, reform has three options: abandon the transplant, adapt the law to local ecology, or transform local ecology to accommodate the law. The third option, ecosystem transformation, is feasible only under exceptional circumstances and requires timescales measured in decades. Most successful transplants either begin with high compatibility or achieve it through extensive local adaptation.

**Anomaly Three: The WEIRD Versus Non-WEIRD Implementation Gap**

The gap between law-on-books and law-in-action exhibits striking, stable disparity across societies. WEIRD societies. Western, Educated, Industrialized, Rich, Democratic, average 5.4% implementation gaps, where implementation gap is defined as difference between formal legal requirements and actual compliance/enforcement. Non-WEIRD societies average 31.2% gaps, a 5.7-fold disparity. This gap has remained stable for four decades (1980-2024) despite massive investments in institutional capacity building, technical assistance, and legal reform. World Bank governance projects invested an estimated $47 billion (2000-2020) in improving legal implementation across developing countries. The gap barely budged.

The pattern exhibits domain specificity. In property law, the WEIRD/non-WEIRD gap is 2.3-fold (relatively modest). In criminal procedure, the gap reaches 13.7-fold (extraordinary). Contract enforcement gaps fall around 3.8-fold. The variation suggests that gap is not general “state capacity” deficit but something more specific to legal domain. Conventional theories struggle to explain this pattern. Capacity-building approaches assume that technical assistance, training, and resource investment will reduce gaps. Yet 40 years of investment produced negligible convergence. GDP-based explanations assume gaps reflect wealth differences. Yet controlling for per capita GDP reduces correlation only modestly (partial $\rho=0.68$, $p<0.001$), wealthy non-WEIRD societies still exhibit large gaps.

Extended Phenotype Theory explains the gap through memetic alignment between legal norms and cultural substrates. WEIRD legal codes coevolved with WEIRD culture over centuries. Joseph Henrich documents how Western culture is unusual, more individualistic, more reliant on abstract rules, more trusting of impersonal institutions, more prone to analytical thinking. These cultural features are both cause and consequence of legal evolution. Western legal systems developed to accommodate and reinforce WEIRD culture. Contract law assumes parties are autonomous individuals entering voluntary agreements. Property law assumes owners are individuals with exclusive rights. Criminal procedure assumes investigation follows rules applied universally. These assumptions are cultural, not universal.

Non-WEIRD cultures operate on different foundations, more collectivist social organization, more embedded in kinship networks, more reliant on personal relationships, more holistic in cognition. Legal systems developed in these cultures historically embedded law in social relationships rather than extracting it into autonomous rules. When non-WEIRD jurisdictions adopt WEIRD legal codes, which most did through colonialism or voluntary transplantation: they import conceptual memes (legal rules) without behavioral memes (the cultural substrate making rules functional).

The result is systematic implementation gaps. Consider contract enforcement. WEIRD contract law assumes parties document agreements formally, interpret documents according to literal text, and enforce through impersonal courts applying universal rules. This system works when parties share these assumptions and possess the literacy, documentation practices, and institutional trust to operationalize them. Non-WEIRD contract behavior often relies on oral agreements, relationship-specific interpretation, and enforcement through social sanction rather than legal proceeding. When WEIRD contract law is transplanted, it creates formal framework at odds with behavioral practice. The gap is not “failure” to implement law properly but rather predictable consequence of importing legal memes without compatible cultural substrate.

Criminal procedure gaps are larger because criminal procedure in WEIRD systems depends heavily on internalized norms, police follow rules even when unobserved, prosecutors respect defendant rights even when inconvenient, judges apply standards consistently even when unpopular. These behaviors require extraordinary levels of impersonal prosociality: treating strangers according to universal rules rather than particular relationships. WEIRD culture cultivates such prosociality through specific historical developments (Catholic Church dissolution of kin networks, Protestant emphasis on individual conscience, market economies rewarding abstract trust). Non-

WEIRD cultures cultivate different prosocial norms, intense loyalty within kinship groups, reciprocity within relationships, face-maintenance in hierarchies. Police, prosecutors, and judges in non-WEIRD societies behave prosocially according to their cultural logic: they protect relationships, fulfill reciprocity obligations, maintain hierarchical order. But this produces outcomes that diverge systematically from WEIRD criminal procedure expectations.

The gap persists because behavioral memes evolve slowly. Changing formal legal codes is relatively easy, a legislature can pass new laws within months. Changing behavioral memes requires transforming the cultural substrate that reproduces behavior across generations. This involves education, media, social practice, and institutional experience sustained over decades. My analysis projects that at current convergence rates (approximately 0.15% annual reduction in gap), full convergence would require 226 years, approximately 11 generations. This is not prediction of actual convergence but rather illustration of timescale mismatch. Legal reform operates on political timescales (years to decades); memetic evolution operates on cultural timescales (generations to centuries).

The three anomalies, transcultural convergence, transplant failure, and the WEIRD gap, share a common structure. Conventional legal theory treats law as representation (of values, functions, or rules) that can be freely transported across contexts. Extended Phenotype Theory treats law as phenotypic machinery embedded in memetic ecology. The machinery succeeds or fails based on compatibility with ecosystem, not inherent quality. Transcultural convergence occurs when different ecologies select for similar phenotypes through identical selection pressures. Transplant failure occurs when phenotypes adapted to source ecology prove maladaptive in recipient ecology. The WEIRD gap persists because WEIRD legal phenotypes require WEIRD cultural ecology to function as designed. In each case, the explanatory power comes from treating law as embodied machinery rather than abstract representation.

## V. THE ULTRAACTIVITY TRAP: WHEN COOPERATION BECOMES STRUCTURALLY DOMINATED

The darkest implication of extended phenotype analysis concerns what I term ultraactivity, institutional features whose benefits, once captured, persist regardless of future cooperation. Ultraactivity transforms repeated democratic games into terminal competitions by collapsing the shadow of the future. The mechanism deserves extended treatment because it explains not merely institutional persistence

but institutional crystallization that makes democratic renovation structurally impossible.

Standard repeated-game theory establishes that cooperation can emerge in infinitely repeated games through the shadow of the future, rational players cooperate today because they expect future interactions and can punish defection later. This logic undergirds democratic politics. Parties alternate in power but maintain constitutional constraints because today's majority will someday be tomorrow's minority, and both sides benefit from preserving rules protecting minority rights. If Party A governs today but faces electoral defeat tomorrow, A refrains from maximally exploiting power because B, when governing, could retaliate by equally exploiting power. Mutual forbearance emerges from iterated interaction.

Ultraactivity breaks this logic. When legal institutions exhibit ultraactivity, benefits captured through defection persist indefinitely, the shadow of the future collapses. Party A can capture benefits during its government that continue even after losing power. Examples include lifetime judicial appointments, irreversible constitutional amendments, permanent fiscal commitments, infrastructure projects locking in relationships. If A can make changes that persist regardless of B's future actions, A's optimal strategy is maximal exploitation rather than mutual forbearance. Recognizing this, B adopts identical strategy when in power. The result is terminal competition, each party exploits power maximally when governing, knowing the other will do likewise, producing ratchet of institutional crystallization.

Argentina demonstrates the mechanism in pure form. The Constitutional Lock-in Index I have developed, which measures institutional rigidity across four dimensions (constitutional text rigidity, judicial doctrine stability, veto actor multiplicity, institutional tenure overlap), assigns Argentina a score of 0.89 on a 0-1 scale where 1 represents maximum rigidity. This score has remained stable across 80 years spanning wildly different regimes, military dictatorships, populist democracies, neoliberal administrations, leftist governments, and libertarian revolution. The score persists because ultraactivity has accumulated across generations through a process I will trace.

The origins lie in corporatist ideology that emerged during the 1940s military government preceding Perón. Two institutional innovations from this period proved decisive: housing regulation (1943) and labor regulation (1953). Both established ultraactive institutions, legal regimes whose benefits to constituencies persisted regardless of future political configuration. Housing regulation created rent controls, tenant protections, and expropriation powers that benefited urban working classes.

Labor regulation created employment stability, union recognition, and extensive workplace protections that benefited organized labor. Both regulations survived Perón's overthrow (1955), demonstrating ultraactivity: they had created constituencies with material interests in persistence and legal infrastructure resisting dismantling.

The critical divergence occurred during the 1976-1983 military dictatorship, which explicitly targeted both regimes for elimination. Housing regulation was successfully abolished in 1976, the military simply repealed tenant protections, eliminated rent controls, and deregulated housing markets. The elimination was permanent; housing regulation never reconstituted itself. Labor regulation, by contrast, survived. The dictatorship attempted elimination repeatedly (1976, 1978, 1980) but each attempt was frustrated by judicial resistance, administrative sabotage, union resilience, and even business preference for maintaining predictable labor relations. By 1983, labor regulation had not only survived but had become template for interpreting all social regulation through framework of "acquired rights" that cannot be diminished.

Why did labor regulation survive when housing regulation did not? The answer is ultraactivity accumulation. Labor regulation had constructed deeper institutional entrenchment through judicial precedent, professional specialization, organizational infrastructure, and constitutional embodiment. When the dictatorship fell (1983), labor forces sought constitutional protection against future assault. They succeeded in embedding Article 14bis into the 1994 constitutional reform, making labor rights constitutional requirements rather than merely legislative policy. This converted ultraactivity from ordinary legislation (reversible by majority) to constitutional provision (reversible only by supermajority under restrictive procedures).

The 1994 constitutional reform proved decisive. Article 14bis not only protects specific labor rights but establishes an interpretive framework treating social rights as irreversible. Courts interpret the provision as prohibiting "regulatory degradation", any legal change diminishing protections violates the Constitution. This ratchets labor protection upward across time. Each new benefit, once granted, becomes constitutional floor. Future governments can add protections but cannot subtract them. The institutional logic is ultraactive in pure form, benefits captured during one government persist regardless of future political composition.

The consequences are visible in reform attempts. Argentina has undertaken 23 documented attempts at labor reform (1983-2025). All 23 failed to produce sustained change. "Failure" here means reforms either blocked before implementation, reversed shortly after implementation, or eviscerated through judicial interpretation while nominally remaining on books. The most recent significant attempt, the Macri

administration's labor flexibility reforms (2017), provides instructive example. After three years of negotiation, coalition-building, and legislative maneuvering, Congress passed modified reforms that modestly relaxed employment termination procedures and union monopolies. Within 18 months, courts had interpreted the reforms to restore previous protections through "constitutional harmonization." Within 30 months, the succeeding Peronist government formally repealed remaining provisions. The reforms achieved zero sustained impact.

The CLI mechanism explains this perfect failure rate. Once an institutional regime achieves CLI above approximately 0.70, reform becomes structurally impossible absent extraordinary crisis sufficient to temporarily suspend normal institutional constraints. Argentina's CLI of 0.89 means that labor institutions have crystallized to near-maximum extent possible within formal democracy. The four components of CLI are each near-maximum: constitutional text protection (0.94), judicial doctrine stability (0.91), veto actor multiplicity (0.85), and institutional tenure overlap (0.86). With scores this high, ordinary political majorities cannot dislodge the regime regardless of electoral mandate, economic necessity, or policy merit.

The Chilean comparison illuminates what low CLI permits. Chile's CLI is 0.24, indicating high institutional flexibility. Chile has successfully implemented labor reforms bidirectionally, increasing protections under left governments (2006-2010, 2014-2018) and reducing protections under right governments (1979-1990, 2010-2014), with 83% of reforms persisting through subsequent government transitions. The difference is not economic development, legal tradition, or even regime type (both countries have been democracies since 1980s). The difference is ultraactivity. Chilean labor law never achieved constitutional status. It remains ordinary legislation adjustable by majority vote. Courts do not interpret employment protection as irreversible constitutional mandate. The labor code is phenotypic machinery, but machinery that remains malleable rather than crystallized.

The ultraactivity trap has profound implications for democratic functionality. When institutions accumulate ultraactive features across generations, democratic alternation becomes progressively more costly. Each party that governs adds irreversible changes benefiting its constituencies. Subsequent parties cannot reverse these changes but can add their own. The institutional environment ratchets toward maximum rigidity. Eventually, institutional space becomes so constrained that incoming governments discover most policy questions have been preemptively decided by previous governments through ultraactive commitments. Democratic choice becomes notional rather than real, voters can change parties but cannot

change policies because policies have been locked in constitutionally, judicially, or administratively.

Argentina is approaching this terminal condition. The current Milei government (2023-present) discovered that most of its proposed reforms violate either constitutional provisions or judicially-created constitutional interpretation or existing legislative framework protected by veto actors. The government can propose change, but effectuating change requires either constitutional amendment (impossible without opposition cooperation given supermajority requirements) or constitutional reinterpretation (which courts have declined to provide) or extended legislative negotiation (which dilutes reforms to symbolic gestures). The space for democratic renovation has collapsed not because voters failed to authorize change. Milei received decisive electoral mandate, but because previous generations used ultraactive mechanisms to preempt future democratic choice.

I am currently monitoring the United States for early signs of similar trajectory. My preliminary analysis suggests USA CLI has increased from approximately 0.33 (1980) to 0.41 (2024): not yet in danger zone but trending poorly. The increase reflects growing judicialization (courts making policy through constitutional interpretation), expanding executive authority (presidents using administrative state to lock in policies), and norm erosion (both parties now willing to exploit temporary advantages through mechanisms that were previously considered out of bounds). If this trajectory continues, projections suggest USA CLI could reach 0.49 by 2035, approaching the critical threshold of 0.50 above which cooperative equilibria become structurally unstable. This is not prediction of inevitable outcome but rather warning based on current trajectory. The USA has stronger tradition of institutional forbearance than Argentina, but traditions can erode rapidly once ultraactive competition begins.

The policy implication is stark: Democracies should prohibit ultraactivity constitutionally. Just as democracies require periodic elections and separation of powers, they should require temporal symmetry: that benefits captured during one government must be reversible by successor governments. This does not mean governments cannot make commitments, only that commitments must remain vulnerable to democratic revision. Lifetime judicial appointments, irreversible constitutional rights, permanent fiscal commitments: these are institutionalized ultraactivity. They may seem to protect valuable principles against democratic whim, but they achieve this by removing principles from democratic contestation altogether. The cumulative effect across generations is democratic exhaustion, the form of democracy persists but its substance has been hollowed out by accumulated ultraactive constraints.

The modern prince learns: Ultraactivity is civilizational trap. The party that introduces ultraactive institutions may gain temporary advantage, but it initiates process that ultimately destroys democratic functionality. Once ultraactivity becomes standard political strategy, all parties adopt it, producing ratchet toward crystallization. The only escape is precommitment to temporal symmetry, constitutional prohibition on institutional changes whose benefits persist regardless of future democratic choice. This requires collective self-restraint rarely observed in competitive politics. But the alternative, demonstrated by Argentina's 80-year trajectory toward institutional paralysis, is democracy reduced to theater where parties alternate in power but policy space has been exhausted by previous generations' ultraactive commitments.

## VI. CONCLUSION: THE PRINCE OF THE 21ST CENTURY

This paper has synthesized Machiavellian political realism with Darwinian evolutionary epistemology to generate a framework for understanding institutional persistence and reform. The synthesis rests on four conceptual innovations: fitness-welfare divergence explains why dysfunctional institutions outcompete functional alternatives; Evolutionarily Stable Strategy analysis clarifies why superior alternatives fail to displace inferior incumbents; extended phenotype theory resolves anomalies puzzling conventional legal theory; and ultraactivity mechanism reveals how temporal asymmetry transforms repeated games into terminal competition.

The implications for governance strategy are profound but require careful statement to avoid naturalistic fallacy. This analysis does not claim that evolution justifies institutional arrangements: "what persists" is not equivalent to "what ought to persist." Rather, evolutionary analysis enables prediction of which institutional arrangements will persist, independent of normative evaluation. A governor who understands memetic fitness, ESS stability, and ultraactivity dynamics can predict reform success or failure with far greater accuracy than conventional political analysis provides. This predictive capacity is valuable not because it justifies current arrangements but because it enables more effective reform strategy when reform is desired.

The difference between Machiavellian political realism and evolutionary institutional analysis parallels the difference between Machiavelli's *Il Principe* and this framework. Machiavelli taught princes to acquire and maintain personal authority through adaptation to circumstance. His prince was individual actor operating within given institutional context, seeking to master that context through *virtù* (skill) despite *fortuna* (chance). Evolutionary analysis teaches governors to engineer institutional

ecosystems that replicate across generations independent of individual governors. The unit of analysis shifts from prince to institution, from personal authority to memetic fitness, from acquisition to reproduction. Where Machiavelli's prince seeks to hold power across his lifetime, the evolutionary governor seeks to construct self-replicating institutional machinery that persists indefinitely.

This shift has ethical implications that must be confronted directly. If institutions can be engineered to persist regardless of welfare outcomes, doesn't this framework provide playbook for entrenching dysfunction? The concern is legitimate but reflects misunderstanding of the framework's purpose. Evolutionary analysis is descriptive: it explains how institutions persist, not whether they should persist. The analysis applies equally to welfare-enhancing and welfare-destroying institutions. Understanding memetic fitness does not require advocating for particular fitness targets any more than understanding human psychology requires advocating for particular psychological states.

Indeed, the framework clarifies ethical responsibility precisely by separating descriptive mechanism from normative goal. A governor who understands fitness-welfare divergence cannot plead ignorance when constructing institutions that optimize for replication rather than welfare. The governor knows -in a way Machiavelli could not- that institutions will optimize for whatever target the selection environment rewards. If the environment rewards short-term constituency mobilization, institutions will evolve to maximize mobilization regardless of governance quality. If the environment rewards long-term welfare production, institutions will evolve to maximize welfare. The governor's responsibility is to shape selection environment to reward desired outcomes, not merely to construct clever institutions.

The ultraactivity analysis adds another dimension of ethical clarity. A governor who understands ultraactivity dynamics cannot honestly claim to "protect" important principles by making them irreversible. Making principles irreversible protects them against future democratic revision, but the cumulative effect across generations is to reduce democratic substance to empty ritual. The protection comes at the cost of democratic vibrancy. This is not merely technical observation but ethical choice. The governor must decide whether protecting particular principles against future democratic choice is worth the long-term democratic exhaustion that accumulated ultraactivity produces. Argentina demonstrates the endpoint of that choice, democracy becomes theater where parties alternate but policy space has been exhausted by previous generations' "protections."

The framework also illuminates limits of evolutionary analysis. Evolution does not select for truth, justice, or beauty: it selects for reproduction. An institution can replicate successfully while being based on falsehoods, producing injustice, and degrading aesthetic value. The governor who understands this cannot confuse evolutionary success with moral worth. The populist strategy's 216-fold fitness advantage over liberal strategy in Argentina does not make populism "truer" or "better", merely more reproductively successful in that ecosystem. Similarly, the labor regime's ultraactive protection does not make it "just" or "necessary", merely successfully self-replicating.

This separability between fitness and worth creates space for conscious choice that pure evolutionary determinism would deny. If institutions were optimizing perfectly for fitness, intervention would be futile: evolutionary pressures would simply override attempts at reform. But institutions optimize imperfectly. There exist opportunities for conscious redesign that can shift institutions toward different equilibria. These opportunities are rare and require precise execution -the ESS analysis made clear that successful transition demands crisis, coalition, speed, and irreversibility simultaneously- but they exist. The New Zealand case demonstrates that transition is possible under proper conditions.

The computational tools developed in my research program -IusMorfos for transplant prediction, CLI for reform capacity measurement, JurisRank for doctrine persistence analysis- operationalize evolutionary analysis in ways that make conscious institutional engineering more feasible. A reformer equipped with these tools can assess ex ante whether proposed reform has reasonable success probability. If IusMorfos Environmental Compatibility score is below 0.65, the reformer knows to expect failure and can either abandon the reform, adapt it to local ecology, or commit to ecosystem transformation before attempting transplant. If CLI exceeds 0.70, the reformer knows that ordinary political means will fail and can either pursue extraordinary means (crisis exploitation, constitutional revolution) or accept that reform is impossible under current conditions.

This predictive capacity transforms political strategy from art to science: or at least from pure art to art informed by science. The governor equipped with evolutionary understanding retains discretion about goals but gains systematic knowledge about means. This parallels how Machiavelli transformed political analysis by introducing empirical observation in place of theological speculation. The moral questions -what goals to pursue, which values to privilege, how to balance competing goods- remain fully open. But the instrumental questions (which reforms will persist, which

institutions will resist change, which strategies have realistic success probability) become answerable through systematic analysis rather than mere intuition.

The concluding implication returns to the title's premise: If Machiavelli had known Darwin, he would have written differently. He would have recognized that power resides not in princely *virtù* but in institutional fitness; that stability depends not on mastering *fortuna* but on constructing self-replicating phenotypic machinery; that acquisition matters less than reproduction. The Prince of the 21st century engineers ecosystems, not personal authority. This prince understands that institutions are memetic replicators subject to selection pressures; that fitness-welfare divergence means successful replication can accompany chronic dysfunction; that ESS stability makes superior alternatives structurally vulnerable to inferior incumbents; and that ultraactivity transforms democratic iteration into terminal competition.

Machiavelli revolutionized political thought by describing power as it operates rather than as it should operate. This framework completes his revolution by explaining *why* power operates as it does: not through appeals to human nature, divine will, or historical accident, but through evolutionary dynamics of variation, selection, and differential reproduction. The framework does not eliminate ethics or reduce politics to biology. Rather, it clarifies the constraints within which ethical political action must operate. The modern prince who ignores these constraints (who assumes welfare merit produces institutional success, that technical quality determines adoption, that good intentions translate into persistent reform) will fail as systematically as Machiavelli's prince who ignored Renaissance political realities. The prince who understands evolutionary dynamics gains no guarantee of success but does gain realistic assessment of what success requires and honest evaluation of whether those requirements can be met.

The path forward is neither evolutionary determinism nor naive voluntarism. Institutions evolve through selection, but selection environment includes conscious human choices about goals, rules, and institutional design. The evolutionary governor shapes that environment deliberately, understanding that institutions will adapt to whatever selection pressures the environment imposes. If we want institutions that promote welfare, we must create environments that reward welfare production. If we want institutions that permit democratic renovation, we must prohibit ultraactivity. If we want pluralism to survive competition with monocultures, we must design protections compatible with pluralism's lower memetic fitness. These are choices, not inevitabilities. But they are constrained choices, bounded by evolutionary dynamics that cannot be wished away. The modern prince learns: Work with evolution, not against it. Design selection environments to reward desired outcomes. Construct

institutional machinery with replication in mind. And avoid ultraactivity, the civilizational trap that sacrifices democratic renewal for temporary advantage. This is not moral philosophy but strategic realism for the 21st century. Machiavelli would approve.

---

## REFERENCES

### Primary Theoretical Sources

Darwin, Charles. 1859. *On the Origin of Species by Means of Natural Selection*. London: John Murray.

Dawkins, Richard. 1976. *The Selfish Gene*. Oxford: Oxford University Press.

Dawkins, Richard. 1982. *The Extended Phenotype: The Long Reach of the Gene*. Oxford: Oxford University Press.

Dennett, Daniel C. 1995. *Darwin's Dangerous Idea: Evolution and the Meanings of Life*. New York: Simon & Schuster.

Henrich, Joseph. 2020. *The WEIRDest People in the World: How the West Became Psychologically Peculiar and Particularly Prosperous*. New York: Farrar, Straus and Giroux.

Kuhn, Thomas S. 1962. *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press.

Machiavelli, Niccolò. 1532 [1961]. *The Prince*. Translated by George Bull. London: Penguin Books.

Maynard Smith, John. 1982. *Evolution and the Theory of Games*. Cambridge: Cambridge University Press.

### Legal and Political Theory

Finnis, John. 1980. *Natural Law and Natural Rights*. Oxford: Oxford University Press.

Hart, H.L.A. 1961. *The Concept of Law*. Oxford: Oxford University Press.

Weber, Max. 1919 [1946]. "Politics as a Vocation." In *From Max Weber: Essays in Sociology*, edited by H.H. Gerth and C. Wright Mills. New York: Oxford University Press.

### Author's Related Publications

Lerer, Ignacio Adrian. 2025a. "Law as Extended Phenotype: Reframing Legal Theory Through Evolutionary Epistemology - Why Legal Positivism and Natural Law Face a Crisis of Anomalies." SSRN Working Paper No. 5737383. Available at: https://ssrn.com/abstract=5737383

Lerer, Ignacio Adrian. 2025b. “The Ultraactivity Trap: How Temporal Asymmetry Transforms Repeated Games into Terminal Betrayal.” SSRN Working Paper No. 5708042. Available at: https://ssrn.com/abstract=5708042

Lerer, Ignacio Adrian. 2025c. “The Extended Phenotype of Populism: A Memetic Analysis of Policy Persistence in Latin America.” SSRN Working Paper No. 5463814. Available at: https://ssrn.com/abstract=5463814

Lerer, Ignacio Adrian. 2025d. “Argentina’s Fiscal Lock-in: Tax Reform as Extended Phenotype.” SSRN Working Paper No. 5635152. Available at: https://ssrn.com/abstract=5635152

Lerer, Ignacio Adrian. 2025e. “Constitutional Paleontology: Tracing the Ancestor’s Tale of Legal Institutions.” SSRN Working Paper No. 5660770. Available at: https://ssrn.com/abstract=5660770

Lerer, Ignacio Adrian. 2025f. “International Law as Extended Phenotype: Globalist and Sovereigntist Memeplexes Competing Through Legal Artifacts (2000-2025).” SSRN Working Paper No. 5612010. Available at: https://ssrn.com/abstract=5612010

Lerer, Ignacio Adrian. 2025g. “Beyond Iusmorphs: Environmental Compatibility Dominates Technical Quality in Legal Transplant Success.” SSRN Working Paper No. 5605730. Available at: https://ssrn.com/abstract=5605730

Lerer, Ignacio Adrian. 2025h. “The Extinction of the Nineteenth Amendment: A Memetic Analysis of Constitutional Mortality.” SSRN Working Paper No. 5599553. Available at: https://ssrn.com/abstract=5599553

Lerer, Ignacio Adrian. 2025i. “Jurists as Evolutionary Engineers: Artificial Selection in Legal Doctrine.” SSRN Working Paper No. 5492066. Available at: https://ssrn.com/abstract=5492066

Lerer, Ignacio Adrian. 2025j. “Why Bad Law Persists: Evolutionary Stable Strategies in Legal Systems.” SSRN Working Paper No. 5478426. Available at: https://ssrn.com/abstract=5478426

Lerer, Ignacio Adrian. 2025k. “The Legal DNA: How Behavioral and Conceptual Memes Shape Institutional Reproduction.” SSRN Working Paper No. 5478467. Available at: https://ssrn.com/abstract=5478467

## Empirical Sources

Bittles, A.H., and Black, M.L. 2010. “Consanguinity, Human Evolution, and Complex Diseases.” *Proceedings of the National Academy of Sciences* 107(supplement 1): 1779-1786.

Murdock, George Peter. 1967. *Ethnographic Atlas*. Pittsburgh: University of Pittsburgh Press.

World Bank. 2024. *World Governance Indicators* (1996-2024). Washington, DC: World Bank Group. Available at: https://info.worldbank.org/governance/wgi/

---

# APPENDIX A: COMPUTATIONAL TOOLS FOR EVOLUTIONARY GOVERNANCE

This appendix provides technical specifications for three computational instruments developed to operationalize Extended Phenotype Theory in governance analysis. All tools are available through open-source repositories for replication and extension.

## A.1 IusMorfos: Legal Transplant Environmental Compatibility Predictor

**Function**: Predicts success probability of legal transplants before implementation by assessing environmental compatibility between source and recipient jurisdictions.

**Accuracy**: 87.4% (validated on 60 transplant cases across Latin America, 2000-2024)

**Methodology**: Multi-dimensional compatibility assessment across five weighted categories

**Formula**:

```
Environmental Compatibility (EC) =
    0.35 × Cultural_Compatibility +
    0.25 × Institutional_Ecology +
    0.20 × Path_Dependency +
    0.15 × Economic_Context +
    0.05 × Technical_Quality
```

**Component Specifications**:

1. **Cultural Compatibility (35% weight)**
    - Regulatory philosophy alignment (cooperative vs. adversarial)
    - Trust in institutions index
    - Formalism vs. flexibility preferences
    - Collectivist vs. individualist orientation
    - *Measurement*: Survey data, comparative legal culture studies, regulatory practice ethnography

2. **Institutional Ecology (25% weight)**
    - Horizontal vs. vertical enforcement capacity
    - Judicial independence vs. political responsiveness
    - Administrative capacity and specialization
    - Professional infrastructure (lawyers, consultants, auditors)
    - *Measurement*: World Governance Indicators, judicial performance metrics, professional density statistics
3. **Path Dependency (20% weight)**
    - Historical precedent in related domains
    - Existing legal framework density
    - Constitutional constraints
    - Vested interests in current arrangements
    - *Measurement*: Legal code analysis, constitutional review, stakeholder mapping
4. **Economic Context (15% weight)**
    - Informal economy size
    - Compliance costs relative to GDP per capita
    - Market structure (competitive vs. concentrated)
    - Enforcement budget availability
    - *Measurement*: National accounts data, regulatory impact assessments, budget analysis
5. **Technical Quality (5% weight)**
    - Legislative drafting clarity
    - Internal consistency
    - International best practice alignment
    - Expert assessment
    - *Measurement*: Comparative legal analysis, OECD regulatory quality indicators

**Interpretation**: - EC ≥ 0.75: High compatibility, full implementation recommended - 0.65 ≤ EC < 0.75: Moderate compatibility, pilot with adaptation recommended - 0.50 ≤ EC < 0.65: Low compatibility, substantial redesign required - EC < 0.50: Incompatible ecology, transplant not recommended without ecosystem transformation

**Example Application**: Argentina Law 27,401 - Cultural Compatibility: 0.22 (adversarial vs. cooperative philosophy) - Institutional Ecology: 0.28 (vertical enforcement high, horizontal weak) - Path Dependency: 0.41 (labor law ultraactivity shapes

interpretation) - Economic Context: 0.35 (informal sector 25% GDP) - Technical Quality: 0.92 (excellent drafting, OECD aligned) - **Overall EC: 0.31 → Prediction: FAILURE** - **Observed Outcome: 3.8% adoption → Prediction validated**

**Replication Repository**: github.com/adrianlerer/IusMorfos - Dataset: 60 transplant cases (2000-2024) - Methodology documentation - Calibration procedures - Analysis scripts (R and Python)

## A.2 CLI: Constitutional Lock-in Index

**Function**: Quantifies institutional rigidity to predict reform capacity and identify crystallized arrangements resistant to democratic renovation.

**Accuracy**: 87% across four Latin American jurisdictions (Argentina, Chile, Peru, Colombia)

**Explained Variance**: $R^2 = 0.84$ (CLI explains 84% of variance in reform success rates)

**Formula**:

```
Constitutional Lock-in Index (CLI) =
    0.30 × CTR +
    0.35 × JDS +
    0.25 × VAM +
    0.10 × ITO
```

**Component Specifications**:

1. **CTR: Constitutional Text Rigidity (30% weight)**
    - Amendment difficulty (supermajority requirements, referendum mandates)
    - Entrenchment clauses (unamendable provisions)
    - Judicial review accessibility and strength
    - Constitutional detail level (specific vs. framework)
    - *Measurement*: Constitutional text analysis, amendment procedure scoring, historical amendment rates
2. **JDS: Judicial Doctrine Stability (35% weight)**
    - Precedent persistence (doctrine survival across Court composition changes)
    - Citation network density
    - Reversal rate (frequency of doctrinal overruling)
    - Interpretive methodology (originalism vs. living constitutionalism)

    - *Measurement*: JurisRank analysis, citation network topology, doctrine genealogy tracing
3. **VAM: Veto Actor Multiplicity (25% weight)**
    - Number of actors with blocking power over reform
    - Geographic distribution (federalism implications)
    - Partisan fragmentation
    - Institutional veto points (bicameralism, executive veto, judicial review)
    - *Measurement*: Political structure analysis, voting threshold requirements, coalition difficulty scoring
4. **ITO: Institutional Tenure Overlap (10% weight)**
    - Personnel continuity across administrations
    - Term length and staggering patterns
    - Removal difficulty for appointed officials
    - Civil service protection strength
    - *Measurement*: Personnel data analysis, statutory term structures, turnover rates

**CLI Scale Interpretation**: - CLI < 0.40: **Flexible** - Bidirectional reforms possible through ordinary political processes - 0.40 ≤ CLI < 0.70: **Moderate Lock-in** - Reforms require extraordinary political effort (crisis + broad coalition) - CLI ≥ 0.70: **High Crystallization** - ESS transition nearly impossible without constitutional rupture

**Validated Cases**:

| Jurisdiction | CLI Score | Reform Success Rate | Pattern |
|---|---|---|---|
| Argentina | 0.89 | 0% sustained (0/23) | Perfect crystallization |
| Chile | 0.24 | 83% sustained (10/12) | High flexibility |
| Peru | 0.52 | 42% sustained (5/12) | Moderate lock-in |
| Colombia | 0.61 | 33% sustained (4/12) | Moderate-high lock-in |

**Component Breakdown - Argentina (CLI = 0.89)**: - CTR: 0.94 (Article 14bis protection, amendment requires 2/3 + constituent assembly) - JDS: 0.91 (labor doctrine 89% citation fidelity across Court changes) - VAM: 0.85 (federal structure + union veto + judicial review + legislative supermajority) - ITO: 0.86 (judges lifetime appointment, union officials rotating but cohesive)

**Component Breakdown - Chile (CLI = 0.24)**: - CTR: 0.22 (simple majority amendments on many issues) - JDS: 0.31 (doctrine reversal accepted, low precedent binding) - VAM:

0.19 (unitary state, few veto points) - ITO: 0.24 (moderate tenure overlap, frequent personnel change)

**Predictive Application**: Before attempting institutional reform, calculate jurisdiction CLI. If CLI > 0.70, conventional reform strategy will fail. Alternative strategies required: - **Crisis exploitation**: Wait for external shock severe enough to temporarily suspend normal constraints - **Incremental mutation**: Target peripheral rather than core institutional features - **Ecosystem transformation**: Change selection environment first, then institutions adapt - **Constitutional moment**: Pursue fundamental regime change rather than within-regime reform

**Replication Repository**: github.com/adrianlerer/law-as-extended-phenotype-kuhn - CLI calculation protocols - Historical reform dataset (23 Argentine attempts documented) - Comparative analysis across jurisdictions - Projection models

## A.3 JurisRank: Judicial Doctrine Fitness Analyzer

**Function**: Measures memetic fitness of legal doctrines through citation network analysis, predicting which doctrines will persist across generational transitions and which are vulnerable to extinction.

**Methodology**: Adapted PageRank algorithm for legal citation networks, incorporating temporal dynamics and citation quality weighting.

**Algorithm**: ABAN (Ancestral Branch Analysis Network) for genealogical doctrine tracing

**Output**: Memetic fitness score (0-1 scale) + inheritance fidelity percentage + vulnerability assessment

**Core Principle**: Doctrines are memes. Citation = reproduction. Network centrality = fitness. High-fitness doctrines resist displacement; low-fitness doctrines face extinction risk.

**Measurement Process**:

1. **Citation Network Construction**
    - Identify doctrine-originating case (root node)
    - Map all subsequent cases citing doctrine (descendant nodes)
    - Weight edges by citation quality:
        - Direct application (weight = 1.0)
        - Interpretive extension (weight = 0.7)

        - Distinguishing citation (weight = 0.3)
        - Critical citation (weight = 0.1)
2. **Fitness Calculation**
    - Apply modified PageRank: $F(d) = (1-\lambda)/N + \lambda \Sigma [F(c) \times w(c \rightarrow d) / C(c)]$
    - Where: $\lambda$ = damping factor (0.85), N = network size, w = citation weight, C = out-degree
    - Normalize to 0-1 scale
3. **Inheritance Fidelity Assessment**
    - Track doctrine formulation consistency across citations
    - Measure semantic drift using textual analysis
    - Calculate fidelity = 1 - (cumulative drift / generations)
4. **Vulnerability Analysis**
    - Citation gap detection (prolonged periods without reproduction)
    - Competing doctrine emergence (alternative formulations gaining fitness)
    - Court composition sensitivity (fitness correlation with personnel changes)

**Example Application: Peralta Doctrine (Argentina CSJN)**

*Background*: Peralta v. Estado Nacional (1990) established extreme deference to executive emergency powers, validating confiscation of bank deposits during hyperinflation.

*Analysis*: - Dataset: 72 CSJN cases (1922-2025) involving executive emergency powers - Network: Peralta root node, 68 descendant citations - Citation quality: 89% direct application, 11% interpretive extension - **Memetic Fitness: 0.87** (high) - **Inheritance Fidelity: 89%** (post-1990, doctrine formulation stable) - **Vulnerability: Low** (consistent citation despite Court composition changes)

*Interpretation*: Peralta doctrine has achieved high fitness through dense citation network and formulation stability. Doctrine persists despite normative controversy because it has embedded itself in institutional infrastructure. Displacement would require either (1) constitutional amendment explicitly limiting emergency powers or (2) prolonged period (15+ years) without crisis triggering doctrine invocation, allowing competing doctrine to establish alternative citation network.

**Contrast: 19th Amendment (United States)**

*Background*: Amendment XIX (1920) guaranteed women's suffrage but generated little subsequent constitutional doctrine.

*Analysis*: - Dataset: 15 Supreme Court cases potentially implicating 19th Amendment (1920-2024) - Network: Amendment as root node, 1 direct citation (Minor v. Happersett precedent), 0 independent doctrine development - Citation gap: 47 years (1973-2020) without meaningful citation - **Memetic Fitness: 0.23** (near extinction) - **Functional Extinction: 93.3%** (14 of 15 cases decided without reference to Amendment)

Interpretation: 19th Amendment achieved its immediate purpose (enfranchisement) but failed to construct phenotypic machinery for self-replication. It generated no independent interpretive doctrine, no specialized judicial expertise, no litigation infrastructure. The Amendment is legally valid but metrically extinct: it persists formally while having no reproductive capacity.

**Applications**:

1. **Reform Strategy**: Target low-fitness doctrines for displacement (easier than high-fitness)
2. **Constitutional Design**: Build citation-generating features into constitutional provisions
3. **Judicial Appointments**: Assess nominees' likely impact on doctrine fitness landscape
4. **Precedent Management**: Identify vulnerable precedents requiring reinforcement
5. **Litigation Strategy**: Frame arguments to leverage high-fitness doctrines

**Technical Requirements**: - Access to jurisdiction's legal database (comprehensive case law) - Citation extraction capability (manual or automated) - Network analysis software (Python NetworkX or R igraph) - Natural language processing for semantic drift measurement

**Replication Repository**: github.com/adrianlerer/peralta-metamorphosis - ABAN algorithm implementation - Peralta doctrine dataset (72 cases) - JurisRank calculation scripts - Visualization tools for citation networks

---

# APPENDIX B: DATASET DESCRIPTIONS

## B.1 Legal Transplant Database (IusMorfos Validation)

**Coverage**: 60 transplant cases across Latin America (2000-2024)

**Source Jurisdictions**: Spain (24 cases), Chile (18 cases), United States (12 cases), Germany (6 cases)

**Recipient Jurisdictions**: Argentina (22 cases), Mexico (14 cases), Colombia (10 cases), Peru (8 cases), Brazil (6 cases)

**Legal Domains**: Corporate law (18 cases), labor law (14 cases), environmental law (12 cases), competition law (10 cases), financial regulation (6 cases)

**Outcome Measurement**: Success defined as achieving ≥60% of source jurisdiction adoption/compliance rates within 5 years of implementation. Binary classification (success/failure).

**Success Rate**: 24% overall (14 successes, 46 failures)

**Key Variables Collected**: - Environmental Compatibility scores (5 dimensions) - Technical quality assessments (expert evaluations) - Implementation strength (enforcement resources, monitoring intensity) - Political support (government commitment, stakeholder alignment) - Economic outcomes (compliance costs, economic impact)

**Availability**: Anonymized dataset available upon request for replication. Full dataset with case identifiers available through institutional review board approved protocols.

## B.2 Constitutional Reform Database (CLI Validation)

**Coverage**: 47 constitutional reform attempts across 4 Latin American jurisdictions (1980-2025)

**Jurisdictions**: Argentina (23 attempts), Chile (12 attempts), Peru (7 attempts), Colombia (5 attempts)

**Reform Domains**: Labor law (18 attempts), fiscal structure (14 attempts), judicial organization (8 attempts), federalism (7 attempts)

**Outcome Measurement**: Success defined as reform persisting through at least one government transition (minimum 4 years) without substantial reversal. Coding based on legislative text analysis, judicial interpretation, and administrative practice.

**Success Rates**: - Argentina: 0% (0/23) - Chile: 83% (10/12) - Peru: 29% (2/7) - Colombia: 40% (2/5)

**CLI Scores**: Argentina 0.89, Chile 0.24, Peru 0.52, Colombia 0.61

**Key Variables**: - CLI component scores (CTR, JDS, VAM, ITO) - Reform characteristics (scope, speed, coalition breadth) - External conditions (crisis presence, economic performance) - Implementation mechanisms (constitutional, legislative, administrative) - Reversal mechanisms (legislative repeal, judicial nullification, administrative erosion)

**Availability**: Full dataset with detailed case documentation available at github.com/adrianlerer/law-as-extended-phenotype-kuhn

## B.3 Implementation Gap Database (WEIRD vs. Non-WEIRD Analysis)

**Coverage**: 193 countries (1980-2024), tracked annually

**Data Source**: World Bank Governance Indicators, supplemented by jurisdiction-specific legal effectiveness studies

**Gap Measurement**: Difference between formal legal requirements (law-on-books) and observed compliance/enforcement (law-in-action), expressed as percentage deviation from full implementation.

**Domain-Specific Measurements**: - Property law: Registration rates, title clarity, enforcement effectiveness - Contract law: Enforcement speed, cost, predictability - Criminal procedure: Rights protection in practice, procedural compliance - Labor law: Regulation compliance, enforcement coverage - Environmental law: Permit compliance, monitoring effectiveness

WEIRD Classification: Based on Henrich (2020) criteria. Western, Educated, Industrialized, Rich, Democratic. 47 countries classified WEIRD, 146 non-WEIRD.

**Key Findings**: - Overall gap: WEIRD 5.4% (±2.1%), Non-WEIRD 31.2% (±18.4%) - Disparity: 5.7-fold, stable 1980-2024 ($r^2$=0.89) - Domain variation: Property 2.3x, Contract 3.8x, Criminal procedure 13.7x - GDP control: Partial correlation ρ=0.68 (gap persists controlling for wealth) - Cultural distance: ρ=0.73 (primary explanatory variable)

**Availability**: Aggregated data publicly available through World Bank. Country-specific detailed analysis available upon request.

### B.4 Transcultural Legal Convergence Database (Extended Phenotype Validation)

**Coverage**: 1,267 societies from Murdock's Ethnographic Atlas, coded for legal/normative patterns

**Primary Variable**: Incest prohibition (parent-child, sibling, extended family)

**Convergence Rates**: - Parent-child + sibling: 99.2% (1,257/1,267 societies) - Parent-child specifically: 99.8% (1,265/1,267) - Sibling specifically: 97.1% (1,231/1,267) - First cousin: 23.4% (296/1,267) [demonstrates selectivity]

**Society Characteristics**: - Geographic distribution: All inhabited continents - Linguistic families: 172 distinct families - Subsistence modes: Foraging, horticultural, pastoral, agricultural, industrial - Political organization: Bands, tribes, chiefdoms, states - Religious systems: Animistic, polytheistic, monotheistic, secular

**Control Analysis**: No historical contact verified for 687 society pairs showing identical prohibition patterns, ruling out diffusion hypothesis.

**Alternative Explanations Tested**: - Functional necessity: Rejected (many societies lack other "functional" norms) - Divine command: Rejected (pattern independent of religious beliefs) - Rational calculation: Rejected (most societies lack genetic knowledge) - Social source: Rejected (no transsocietal legal authority)

**EPT Explanation Validated**: Demographic selection at population level (inbreeding depression coefficient $\beta \approx 0.27$ provides sufficient fitness penalty for selection to operate across ~500 generations).

**Availability**: Murdock's Ethnographic Atlas publicly available. Extended analysis with demographic modeling available at github.com/adrianlerer/LegalGapDB

---

## APPENDIX C: REPLICATION MATERIALS AND OPEN-SOURCE REPOSITORIES

All computational tools, datasets (where permissible), and analysis scripts are available through open-source repositories to facilitate replication, extension, and criticism.

## C.1 Primary Repository: law-as-extended-phenotype-kuhn

**URL**: github.com/adrianlerer/law-as-extended-phenotype-kuhn

**Contents**: - **README.md**: Comprehensive project overview - **REPLICATION_GUIDE.md**: Step-by-step replication instructions - **data/**: Links to all datasets (direct hosting where permissible, access protocols where restricted) - **tools/**: CLI calculator, IusMorfos framework, JurisRank implementation - **analysis/**: R and Python scripts for all statistical analyses - **docs/**: Detailed methodology documentation - **paper/**: Working paper PDF and supplementary materials

**License**: MIT License (tools and code), Creative Commons BY 4.0 (documentation and datasets where authored)

## C.2 Specialized Repositories

**peralta-metamorphosis** - URL: github.com/adrianlerer/peralta-metamorphosis - Contents: JurisRank tool, RootFinder genealogy analysis, Peralta doctrine dataset (72 CSJN cases 1922-2025) - Purpose: Judicial doctrine fitness analysis replication

**nineteenth-amendment-extinction** - URL: github.com/adrianlerer/nineteenth-amendment-extinction - Contents: RootFinder citation analysis, functional extinction documentation, comparative analysis with 2nd Amendment - Purpose: Constitutional mortality case study replication

**LegalGapDB** - URL: github.com/adrianlerer/LegalGapDB - Contents: 19 validated Argentine implementation gap cases, cross-national comparative analysis (193 countries) - Purpose: WEIRD/non-WEIRD disparity analysis replication

**RootFinder-Production** - URL: github.com/adrianlerer/RootFinder-Production - Contents: ABAN algorithm implementation, policy genealogy tracing methodology, visualization tools - Purpose: General-purpose institutional genealogy analysis

## C.3 Data Access Protocols

**Publicly Available**: - CLI scores (4 jurisdictions) - IusMorfos validation results (anonymized) - Ethnographic Atlas incest prohibition data - World Bank Governance Indicators - Constitutional text analysis

**Available Upon Request** (academic use): - Full transplant database with case identifiers (requires IRB protocol) - Detailed reform attempt coding (Argentina 23 cases) - Interview data from regulatory agencies (confidentiality agreements apply)

**Proprietary/Restricted**: - Some judicial databases require institutional access - Certain country-specific enforcement data subject to non-disclosure agreements - Business compliance data aggregated to protect firm confidentiality

## C.4 Software Dependencies

**R Packages** (version 4.3.0+): - igraph (network analysis) - dplyr, tidyr (data manipulation) - ggplot2 (visualization) - lme4 (mixed effects models)

**Python Libraries** (version 3.10+): - NetworkX (network analysis) - pandas, numpy (data manipulation) - scikit-learn (machine learning) - matplotlib, seaborn (visualization)

**Complete dependency lists and installation instructions in repository README files.**

## C.5 Computational Requirements

**Minimal Configuration** (IusMorfos, CLI calculation): - Standard laptop (8GB RAM, dual-core processor) - Execution time: <5 minutes per analysis

**Recommended Configuration** (JurisRank, large-scale analysis): - 16GB+ RAM (for large citation networks) - Multi-core processor (parallel processing) - Execution time: 30 minutes to 2 hours depending on network size

**High-Performance Configuration** (cross-national comparative analysis): - 32GB+ RAM - GPU acceleration (for NLP tasks) - Execution time: 2-8 hours for complete replication

## C.6 Contact and Collaboration

**Author Contact**: adrian@lerer.com.ar

**Collaboration Opportunities**: - Cross-jurisdictional validation (extending beyond Latin America) - Domain-specific applications (corporate law, environmental law, human rights) - Computational tool refinement and extension - Dataset expansion with new cases and variables - Replication in non-legal institutional domains

**Citation Request**: Researchers using these tools or datasets in publications should cite both relevant working papers and the specific repository used. Standard citation formats provided in repository README files.

---

## AI ASSISTANCE ACKNOWLEDGMENT

This paper's core theoretical framework, conceptual innovations, and empirical findings originated from the author's independent research program across multiple SSRN publications (2024-2025). The fitness-welfare divergence principle, ultraactivity trap mechanism, and Extended Phenotype Theory of law all emerged from the author's analysis of institutional persistence patterns.

Artificial intelligence tools (large language models) provided substantial assistance in three areas: (1) mathematical formalization, translating institutional dynamics into quantifiable metrics (IusMorfos formula, CLI components, JurisRank algorithm); (2) literature integration, synthesizing connections between comparative institutional analysis and evolutionary theory (Dawkins, Dennett, Henrich); (3) exposition, organizing complex arguments into coherent narrative structure. AI assistance did not generate empirical findings, design the research program, or formulate core theoretical claims.

The Constitutional Lock-in Index, ultraactivity identification, and Law 27,401 failure prediction all derived from domain expertise and causal inference, not algorithmic analysis. All empirical claims remain the author's sole responsibility and are verifiable through cited sources.

All theoretical innovations are defended as original contributions to evolutionary governance theory. This methodology leverages computational tools for precision and synthesis while retaining human judgment for domain expertise and theoretical innovation.